PROJECT 2.3 · UNIT 2 — MECHANISMS & MOTION

Windmill Construction

Student Edition — VEX EXP · C++ in VS Code

Name: ______________________________ **Date:** ____________ **Period:** ________

Introduction

Your first real engineering **project** — a problem to solve, not a worksheet with an answer key. A **windmill** captures rotary motion and turns it into useful work; your job is to build a geared windmill that **lifts a small load**. You'll run the full engineering design process — define, brainstorm, select with a **decision matrix**, build, test, and improve — and put Unit 2 to work: a **bevel gear** to turn the corner 90° and a **gear ratio** chosen for enough torque to lift. Document every step in your engineering notebook. Read Chapter 5 in the textbook for the full explanation.

Learning Objectives

- Apply the engineering design process from problem definition through iteration.
- Use a decision matrix to choose among design ideas on purpose, not by gut feeling.
- Use a bevel gear to change the direction of motion 90°, and select/justify a gear ratio for the lifting task.
- Document the process thoroughly in an engineering notebook.

Key Ideas (quick reference)

- **The design loop:** define → brainstorm (≥3 sketches) → select (decision matrix) → build → test → improve, then loop. Documenting in the notebook runs through every step. Cycling Test → Improve → Build several times is normal.
- **Decision matrix:** list criteria, weight each 1–5 by importance, score each concept 1–5, then score × weight and total the columns. The highest total is your starting design — and the math shows which features were worth it.
- **Turn the corner:** a bevel gear transfers rotation 'around the corner,' changing the axis of motion 90° — the windmill's signature mechanism.
- **Gear for lift:** lifting takes torque. If it spins free but stalls under load, gear DOWN (slower drum, more pull). The trade is speed — for lifting, slower-but-stronger wins.

Build Constraints

> **Build constraints (all year):** (1) Build from your classroom Gateway kit — including the bevel gears (Advanced Gear Kit) and, for the lift, the Winch & Pulley Kit. It's mostly EXP plus some V5 mechanism parts, all yours to use. Just don't pull from the separate competition team's V5 kit, so it stays complete. (2) The whole windmill must fit one storage cubby — an IKEA Kallax shelf compartment: a 33 × 33 cm (13 × 13 in) opening, about 38 cm (15 in) deep. The 33 × 33 cm face is the binding limit (the build must pass through it). Sketch with that envelope in mind, and break the build down to store it.

Safety

- Keep fingers, hair, and loose clothing clear of meshing gears and the spinning rotor — gear teeth pinch.
- The lifted load can swing or drop: keep it light, lift it over the table (not over hands or laps), and

stand clear of the drop zone.
- If a motor drives the rotor, clear the gear train before it spins and stop it before adjusting anything.

Equipment & Materials

- VEX EXP kit — bevel gears (Advanced Gear Kit) and spur gears (24/36/48/60T)
- A gearbox bracket to hold the bevel pair in proper 90° mesh
- Shafts, bearing flats, shaft collars, C-channels/plates for a rigid frame
- String and a small drum/spool; the standard load your teacher provides
- Design brief and decision-matrix templates (below); engineering notebook
- Optional: an EXP Smart Motor or hand crank to drive the rotor; a fan for true 'wind'

Procedure

1. **Define.** Read the design brief and restate the problem and constraints in your notebook, in your own words.
2. **Brainstorm.** Sketch at least three windmill concepts. Label the input (rotor) and output (lifting drum) on each, and note how each turns the corner 90° and where its gearing goes.
3. **Select.** Fill in the decision matrix below (criteria, weights, scores), total the columns, and choose your design. Record the choice.
4. **Build.** Construct your windmill from EXP parts. Keep the frame rigid so the gears stay meshed, and confirm it fits the cubby.
5. **Model the three gearings.** Set up A (1:1), then B (geared down for torque), then C (geared up for speed). Name the mechanism in each and record what happens to the lift.
6. **Test & improve.** Try to lift the load. If it stalls, gear down further or stiffen the structure; loop until it lifts reliably, recording each change.
7. **Gallery walk.** Compare with other teams' windmills; give and collect peer feedback in your notebook.

Design Brief — restate it in your own words

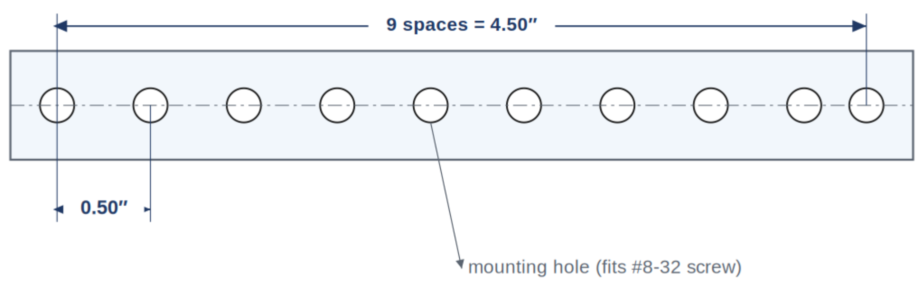
Design brief	Your restatement
Client	*Your teacher (someone who needs a load lifted).*
Problem statement	
Design statement	
Constraints	*classroom Gateway kit · fits the Kallax cubby (33 × 33 cm opening) · changes direction 90° · stable & rigid*
Deliverables	*Working windmill · completed decision matrix · documented notebook*

Decision Matrix

Weight each criterion 1–5, score each concept 1–5, then write **score → score × weight**. Total the columns; the highest total is your starting design.

Criterion	Weight (1–5)	Concept A (score → ×wt)	Concept B	Concept C
Lifting power (torque)				
Turns the corner 90°				
Stable & rigid				
Easy to build				
Fits the cubby				
Total				

Engineering Drawing



RULE: VEX holes are 1 pitch = 0.50″ apart, center-to-center. Beam length = (number of spaces) × 0.50″.

A VEX beam, dimensioned. Holes are one pitch (0.50 in) apart, center-to-center — count the spaces to get the length, and to check the cubby fit.

Conclusion Questions

Answer in complete sentences.

1. How did your gear ratio choice affect the windmill's ability to lift the load?
2. What would you change if you built it again, and why?
3. Which mechanisms from Unit 2 can increase torque and decrease speed? (List all that apply.)
4. Which mechanisms can transmit rotary motion at a 90° angle?
5. What is the purpose of an idler gear — and did your windmill use one?
6. How did the decision matrix change (or confirm) the design you would have picked by gut feeling?